VELS
INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Estd. u/s 3 of the UGC Act, 1956)
PALLAVARAM, THALAMBUR, PERIYAPALAYAM-CHENNAI
ACCREDITED BY NAAC WITH 'A++' GRADE



DERMA VISION AI – A REGION BASED MULTI MODEL SYSTEM FOR SKIN DISEASE DETECTION WITH AI-GENERATED PATIENT GUIDANCE

THIRD REVIEW

Team Members:

Kishore S (23619214)
MohammadUsman(23619222)
AnthonyBenny A(23619202)

Team Guide:

Mrs. Shanthi P
Assistant Professor

Department of Artificial Intelligence and Machine Learning

PROBLEM STATEMENT:

- Skin diseases affect a large population, but dermatologists remain scarce, especially in rural regions, delaying diagnosis and treatment.
- The base paper (SkinGPT-4) uses one single model for all body parts, so it frequently misclassifies diseases that look similar across different regions.
- Existing systems don't verify whether the uploaded image actually matches the selected body part, forcing out a diagnosis even on wrong images.
- AI explanations can hallucinate – generating confident but false medical information, which is risky in a healthcare context.
- Most tools stop at diagnosis and offer no next steps like nearby clinics or consultation tracking.

Our Solution:

•Replace the single general-purpose model with 7 region-specific models, each trained only on its own body part for better accuracy.

•Add a rejection mechanism that checks image-region match before any diagnosis is generated.

•Use a RAG pipeline to verify predictions against a trusted medical knowledge base before the LLM writes an explanation, cutting down hallucination.

•Extend the system beyond diagnosis with self-care tips, consultation reminders, a clinic locator, and secure login.

OBJECTIVES

- Design region-specific deep learning models to improve classification accuracy per body part.
- Build an image-region validation step so mismatched uploads are caught, not diagnosed.
- Ground all LLM explanations in a verified knowledge base to reduce hallucinated medical claims.
- Reach 93–96% classification accuracy, benchmarked against the SkinGPT-4 base paper's reported performance.
- Deliver plain-language diagnosis, causes, self-care guidance, and a mandatory doctor-consultation prompt.

LITERATURE SURVEY :

1 (Base)	"Pre-trained multimodal large language model enhances dermatological diagnosis using SkinGPT-4" – Zhou, J. et al., *Nature Communications*, 15, 5649	Published 05 July 2024	ViT+Llama-2-13b; 80.63% agreement, 93.73% consistency	Single model for all regions; no rejection check
2	"Interpretability of a Deep Learning Based Approach for the Classification of Skin Lesions into Main Anatomic Body Sites" – Jaworek-Korjakowska, J. et al., *Cancers*, 13(23), 6048	Published 01 December 2021	Transfer learning, 3 regions; ~91.45% accuracy	Only 3 regions; no rejection mechanism
3	"DermAI: An Innovative AI-Driven Chatbot for Enhanced Dermatological Diagnosis and Patient Interaction" – Rajeshkumar, P. et al., *IJEEI*, 12(4)	Published December 2024	ResNet50 + RAG chatbot; ~92.6% accuracy	One model for all regions

LITERATURE SURVEY (CONTINUE)

4	"ScalpEye: A Deep Learning-Based Scalp Hair Inspection and Diagnosis System for Scalp Health" – Chang, W.-J. et al., *IEEE Access*, 8, 134826–134837	Published 21 July 2020	Faster R-CNN, scalp-only; 97–99% precision	Single region only
5	"Developing and evaluating a mobile-based teledermatology system for skin lesion diagnosis in Iran" – Shajirat, Z. et al., *Scientific Reports*, 14, 20320	Published 02 September 2024	Teledermatology app, human-reviewed	No AI diagnosis at all
6	"An Integrated Deep Learning Model with EfficientNet and ResNet for Accurate Multi-Class Skin Disease Classification" – Alruwaili, M. & Mohamed, M., *Diagnostics*, 15(5), 551	Published 25 February 2025	EfficientNet+ResNet fusion; ~94%+ accuracy	No region specialization

PROPOSED SYSTEM

- 7 independently trained region models covering Face, Hands, Legs, Feet, Back, Scalp, and Whole Body.
- Each model includes a built-in rejection check to catch region-mismatched uploads before prediction.
- Predictions pass through a RAG pipeline that cross-checks them against verified medical data before explanation generation.
- Final output includes diagnosis, cause, self-care guidance, a doctor-consultation

SYSTEM ARCHITECTURE :

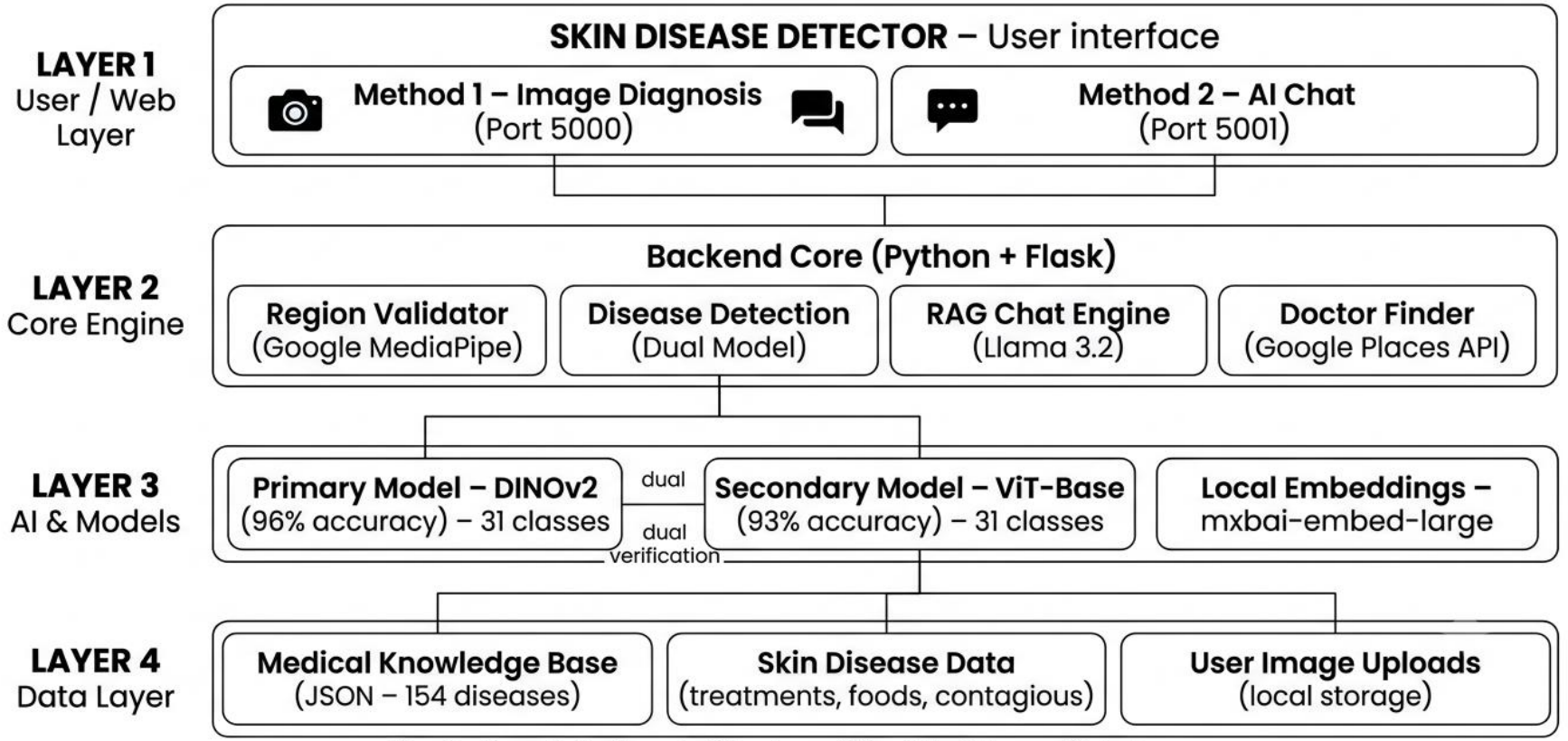
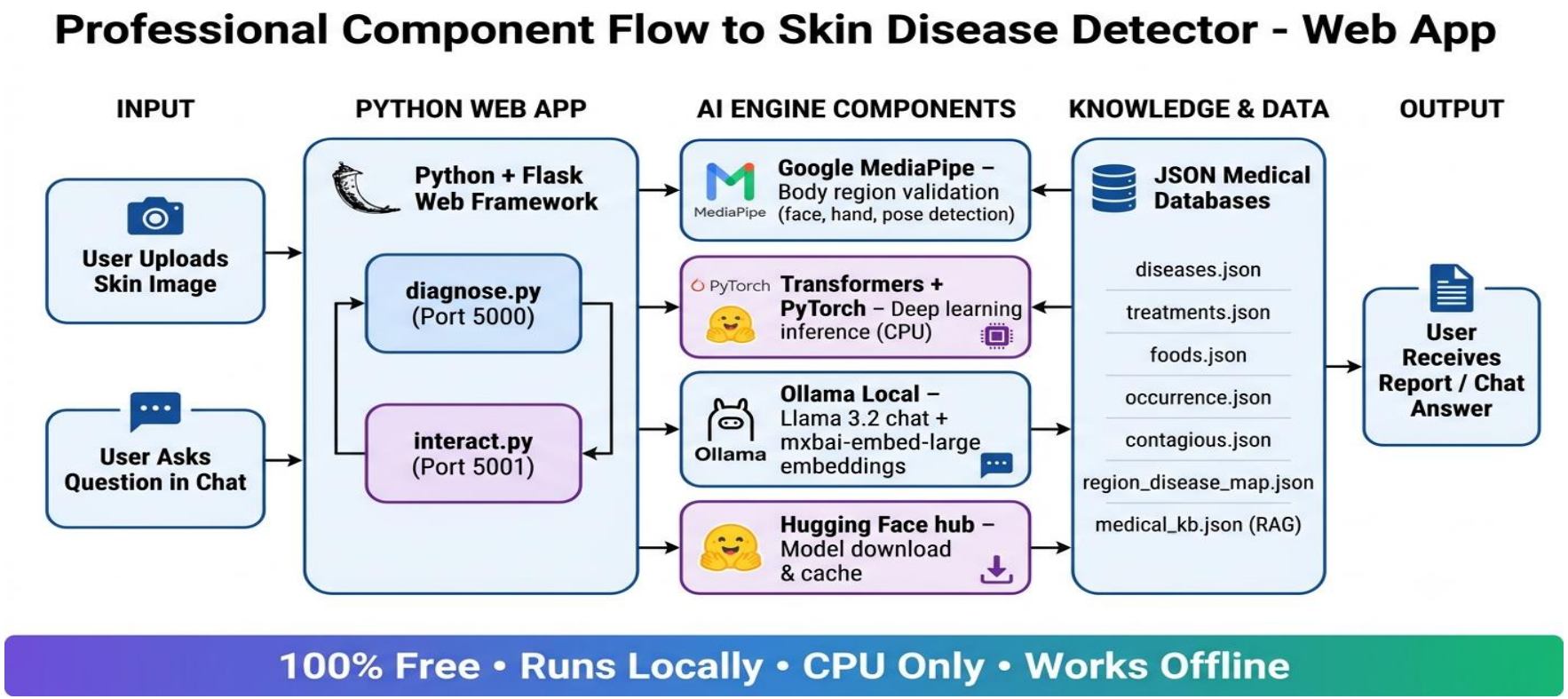


Fig. 1. Overall System Architecture of the Skin Disease Detector.

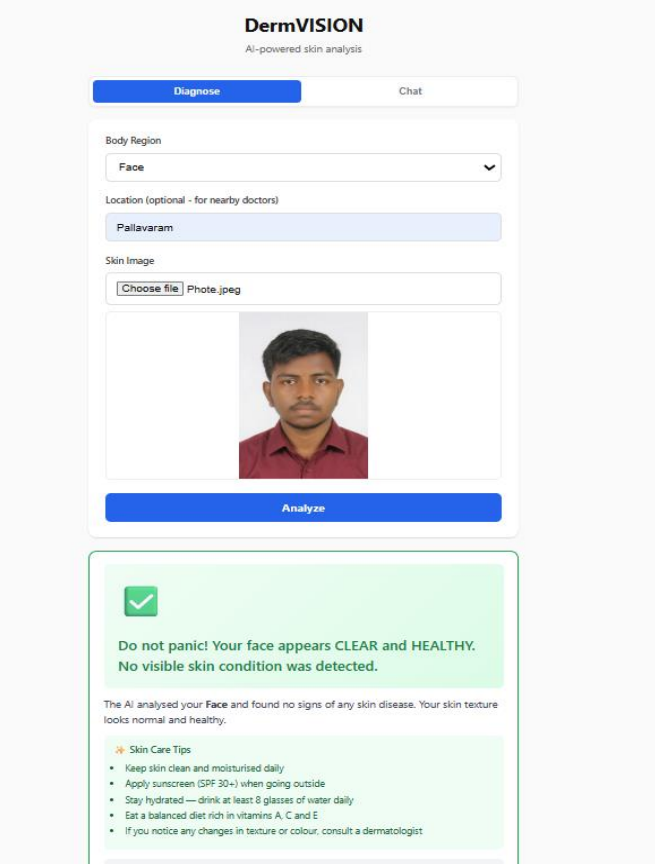
FLOW DIAGRAM



SYSTEM MODULES

- Client Layer: web/app interface with Google OAuth login for secure access.(not Implements yet)
- Image Upload + Region Selector: user uploads a photo and picks the relevant body region.
- Region-Specific Model Layer: 7 CNN models, each specialized for one body part.
- Rejection Mechanism: flags images that don't match the selected region before prediction.
- Prediction Layer: outputs disease name with a confidence score.
- RAG Knowledge Layer: retrieves verified medical facts via **Ollama mxbai-embed-large** to check the prediction.
- LLM Explanation Engine: converts the verified result into a plain-language report.
- Output Layer: doctor-consultation reminder plus a geolocation-based clinic finder.

IMPLEMENTATION -



CHALLENGES FACED

- Early self-trained models had very low accuracy due to insufficient region-specific training data.
- Teaching the system to reliably detect a wrong body-part upload took extra engineering beyond basic classification.
- With no "healthy skin" class in training data, a confidence-threshold fallback had to be built to say "skin looks clear."
- Unstable internet caused repeated failed model downloads, requiring a resume-and-retry mechanism.
- No GPU access meant optimizing models to run efficiently on CPU alone.
- Keeping the entire system free, local, and offline added constraints on model size and hosting choices.
- Available datasets lack diversity in skin tone and lighting, limiting real-world generalization.

SOFTWARE REQUIREMENTS

•**Languages**: Python, JavaScript, HTML, CSS.
• **ML Stack**: Hugging Face Transformers, PyTorch (CPU-optimized), fined-tuned DINOv2 + ViT models, MediaPipe (Google) for body-region detection.
•**RAG/Vector DB**: Ollama mxbai-embed-large for embeddings + local JSON knowledge base (medical_kb.json) with cosine-similarity retrieval.
•**Backend**: Flask Framework.
• **Frontend**: HTML/CSS/JavaScript.
• **Database**: JSON knowledge files (diseases, treatments, foods, occurrence, contagious, region map) — fully local; Neon PostgreSQL (planned future addition).
• **Authentication**: Not needed (100% local app); Google OAuth with JWT (planned future) for any online version.
• **Models**: Jayanth2002/dinov2-base-finetuned-SkinDisease (95.57%) + Jayanth2002/vit_base_patch16_224-finetuned-SkinDisease (93.43%), 31 classes each.
- **Tools**: Git/GitHub for version control; Ollama CLI; Hugging Face Hub; Google Maps Places API + OpenStreetMap for doctor search; Render/Vercel for future deployment.

HARDWARE REQUIREMENTS

•**Processor** (CPU): Dual-core 2.0 GHz minimum; 4+ core (Intel i5 / AMD Ryzen 5) recommended for smooth dual-model inference.

•**RAM**: 6 GB minimum; 8–16 GB recommended (both AI models load together).

•**GPU**: Not required — CPU-only inference; optional NVIDIA GPU speeds up results.

• **Storage**: 3 GB free minimum (models ≈ 1.2 GB + Python packages ≈ 2 GB).

• **Network**: Internet only for one-time setup (download models + packages); fully offline afterward.

• **OS**: Windows 10/11 (also Linux/macOS compatible).

THANK YOU !!